Understanding DNA Damage and Repair

Case Study: Sickle Cell Disease and the HBB Gene

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1. Introduction

Sickle cell disease (SCD) is an inherited blood disorder caused by a single-letter change in the DNA of the *HBB* gene, which codes for a part of haemoglobin. It is inherited in an autosomal recessive manner: people with two copies of the altered gene (HbSS) have sickle cell anaemia, while carriers with one copy (sickle cell trait) are usually healthy. SCD is common in sub-Saharan Africa, India, the Middle East and the Mediterranean, and more than 300,000 babies are born with it every year [1]. The high frequency is linked to malaria, because carriers are partly protected against severe malaria. SCD is a classic example of how one small DNA change can alter a protein and produce a serious disease.

2. Normal Gene and Protein

- **Gene name:** *HBB* (haemoglobin subunit beta), located on the short arm of chromosome 11 (11p15.4) in the β-globin gene cluster.
- **Protein produced:** β-globin, a chain of 146 amino acids. Two β-globin chains combine with two α-globin chains and four iron-containing haem groups to form adult haemoglobin (HbA, $\alpha_2\beta_2$).
- **Normal function:** Haemoglobin inside red blood cells picks up oxygen in the lungs and releases it in the tissues. Normal HbA stays dissolved even after it releases oxygen, so red blood cells remain soft, flexible, disc-shaped cells that can squeeze through narrow capillaries [1].

3. Mutation

A **mutation** is a permanent change in the DNA sequence. Mutations can arise from copying errors during DNA replication or from DNA damage that was not repaired correctly. They range from single-base changes (point mutations) to insertions, deletions and large chromosomal changes [2].

- **Type of mutation:** a *point mutation* – specifically a single-base *substitution* (A→T, a transversion) that changes one amino acid, i.e. a *missense* mutation.
- **Where it occurs:** in codon 6 of exon 1 of the *HBB* gene. The codon GAG (glutamic acid) becomes GTG (valine). This is written as Glu6Val (or p.Glu7Val in modern nomenclature, which counts the starting methionine). It is a germline mutation, so it is present in every cell and passed from parents to children [1].

4. Effect on Protein

The change follows the central dogma, DNA → RNA → protein, as summarised in Table 1 and Figure 1.

Table 1. How one base change in *HBB* changes the protein

Level	Normal (HbA)	Sickle (HbS)
DNA (codon 6)	GAG	GTG
mRNA	GAG	GUG
Amino acid 6	Glutamic acid (charged, water-loving)	Valine (uncharged, water-repelling)
Protein behaviour	Stays dissolved when deoxygenated	Sticks to other HbS and forms long polymers when deoxygenated
Red blood cell	Flexible disc, lives ~120 days	Rigid sickle shape, lives ~10–20 days

Source: compiled by the author from Kato et al. (2018) [1].

Valine is hydrophobic, so when HbS releases oxygen, the valine on one haemoglobin molecule fits into a pocket on a neighbouring molecule. The molecules link into long, stiff fibres that bend the red blood cell into a sickle shape. Sickled cells break down early (**haemolytic anaemia**, causing tiredness and jaundice) and get stuck in small blood vessels (**vaso-occlusion**), causing painful crises, stroke, acute chest syndrome, spleen damage and long-term organ damage [1].

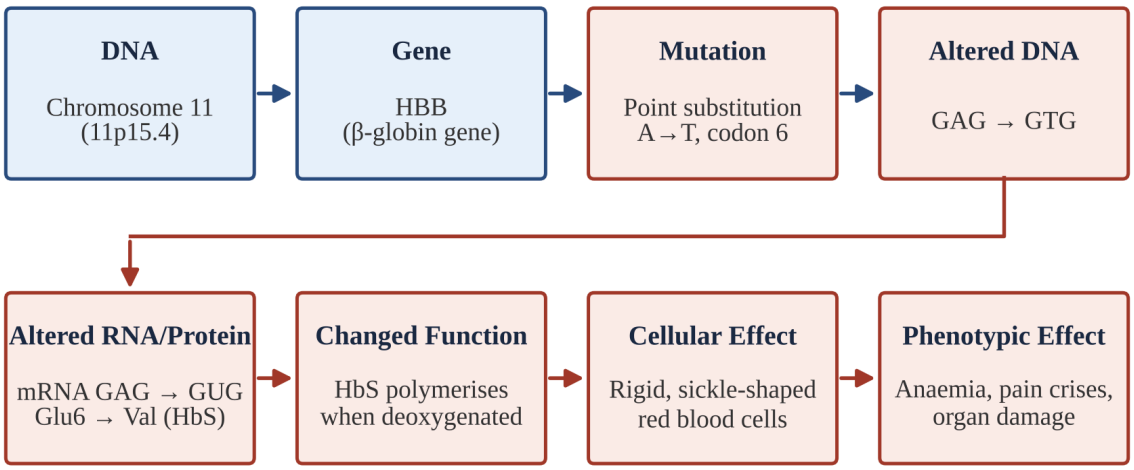


Figure 1. From DNA to phenotype in sickle cell disease. Original flowchart by the author; content based on [1].

5. DNA Repair

Why DNA gets damaged: DNA is damaged all the time. Internal sources include replication errors, reactive oxygen species from normal metabolism, and spontaneous loss or chemical change of bases. External sources include UV light (which joins neighbouring pyrimidines into dimers), ionising radiation (which can break both strands) and chemicals such as tobacco-smoke carcinogens. Each human cell can suffer tens of thousands of DNA lesions per day [2].

Why repair is necessary: Unrepaired damage can block replication and transcription, kill cells, or become fixed as a permanent mutation when the DNA is copied. Mutations in key genes can cause inherited disease, cancer and ageing. Once a change such as the sickle mutation is copied into both strands, repair enzymes can no longer recognise it as damage – it

simply becomes the new sequence. This is why catching damage *before* replication is so important [2].

Main DNA repair pathways (basic level) [2, 3]:

- **Base Excision Repair (BER):** removes a single damaged or wrong base (e.g. an oxidised or deaminated base), then fills and seals the gap.
- **Nucleotide Excision Repair (NER):** cuts out a short stretch of DNA containing a bulky, helix-distorting lesion such as a UV-induced thymine dimer. Defects cause xeroderma pigmentosum.
- **Mismatch Repair (MMR):** corrects wrongly paired bases left behind by DNA polymerase during replication. Defects (e.g. in *MLH1* or *MSH2*) cause Lynch syndrome.
- **Homologous Recombination (HR):** repairs double-strand breaks accurately by copying the matching sequence from the sister chromatid. Requires BRCA1, BRCA2 and RAD51.
- **Non-Homologous End Joining (NHEJ):** repairs double-strand breaks by directly joining the broken ends without a template. Fast but can add or lose a few bases.

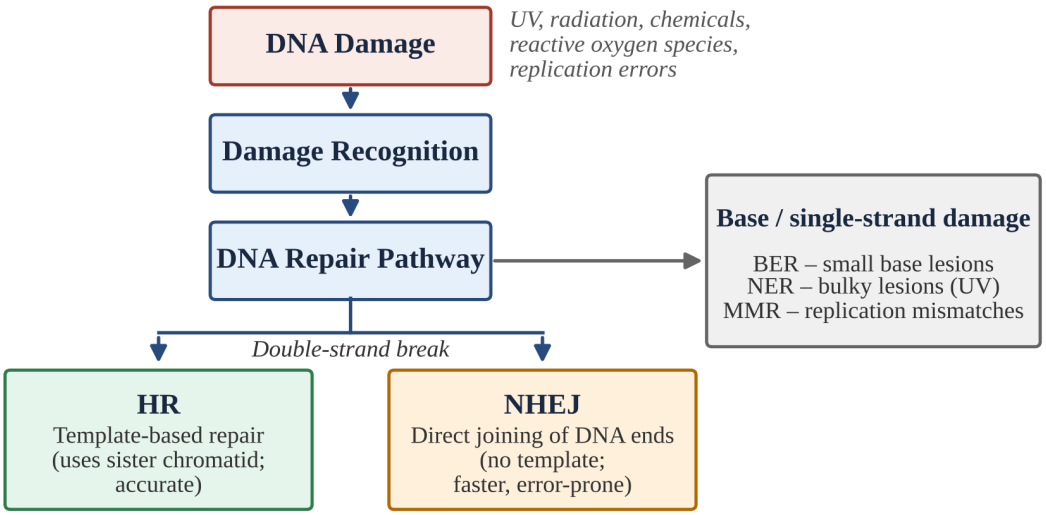


Figure 2. General DNA damage-response and repair pathway choice. Original flowchart by the author; content based on [2, 3].

6. HR vs NHEJ

Both HR and NHEJ repair double-strand breaks, the most dangerous form of DNA damage, but they work very differently (Table 2).

Table 2. Comparison of Homologous Recombination and Non-Homologous End Joining

Feature	Homologous Recombination (HR)	Non-Homologous End Joining (NHEJ)
Damage repaired	Double-strand breaks	Double-strand breaks
Template needed?	Yes – identical sister chromatid	No – ends joined directly
Cell-cycle phase	S and G2 only (after DNA is copied)	All phases; main pathway in G1

Feature	Homologous Recombination (HR)	Non-Homologous End Joining (NHEJ)
Accuracy	High (error-free)	Lower (error-prone; small insertions/deletions)
End processing	Extensive: 5′ ends trimmed to form 3′ single-strand tails	Minimal trimming before ligation
Speed	Slower	Faster
Key proteins	MRN complex, BRCA1, BRCA2, RAD51	Ku70/Ku80, DNA-PKcs, XRCC4–DNA ligase IV
If defective	Breast/ovarian cancer risk (BRCA1/2)	Radiation sensitivity, immune deficiency

Source: compiled by the author from Scully et al. (2019) [3] and Chatterjee & Walker (2017) [2].

These pathways also matter for treating SCD. The CRISPR-based therapy exagamglogene autotemcel (Casgevy), approved in 2023, cuts the patient's stem-cell DNA at a regulator of the *BCL11A* gene. The cell repairs the cut mainly by NHEJ, whose small errors disable the regulator and switch fetal haemoglobin back on, which prevents sickling [4]. Approaches that aim to correct the GTG codon itself instead depend on HR-type, template-based repair.

7. Conclusion

Sickle cell disease shows clearly how **DNA integrity → gene sequence → protein → cellular function → phenotype** are linked: a single A→T change in *HBB* swaps one amino acid in β-globin, makes haemoglobin polymerise, deforms red blood cells and produces a lifelong disease. Because DNA is damaged constantly, cells rely on BER, NER, MMR, HR and NHEJ to fix errors before they become permanent mutations. Without these repair systems, harmful changes would build up in every generation of cells, leading to genetic disease, cancer and loss of genome stability. Understanding these pathways is also opening the way to gene-editing cures that use the cell's own repair machinery.

References

[1] Kato, G. J., Piel, F. B., Reid, C. D., et al. (2018). Sickle cell disease. *Nature Reviews Disease Primers*, 4, 18010. https://doi.org/10.1038/nrdp.2018.10

[2] Chatterjee, N., & Walker, G. C. (2017). Mechanisms of DNA damage, repair, and mutagenesis. *Environmental and Molecular Mutagenesis*, 58(5), 235–263. https://doi.org/10.1002/em.22087

[3] Scully, R., Panday, A., Elango, R., & Willis, N. A. (2019). DNA double-strand break repair-pathway choice in somatic mammalian cells. *Nature Reviews Molecular Cell Biology*, 20(11), 698–714. https://doi.org/10.1038/s41580-019-0152-0

[4] Frangoul, H., Altshuler, D., Cappellini, M. D., et al. (2021). CRISPR-Cas9 gene editing for sickle cell disease and β-thalassemia. *New England Journal of Medicine*, 384(3), 252–260. https://doi.org/10.1056/NEJMoa2031054